

CoRoT-1b

R-2399

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_220106 · created 2026-08-21T00:20:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459585.6999 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.04
Transit depth (Rp/Rs)^2	2.02 +/- 1.14 [%]
Orbital Inclination (inc)	88.53 +/- 2.73
Airmass coefficient 1 (a1)	1.124 +/- 0.027
Airmass coefficient 2 (a2)	-0.096 +/- 0.045
Scatter in the residuals of the lightcurve fit is	2.38 %
Best Comparison Star	None
Optimal Aperture	3.07
Optimal Annulus	11.57
Transit Duration (day)	0.0599 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.04 vs 0.1388 (deviation 0.08 σ)
Duration fit vs published	0.0599 d vs 0.1046 d (deviation 7.45 σ)
Mid-time O-C	-6.1 min
Depth SNR	3.5
Residual scatter	2.38 %

Light curve

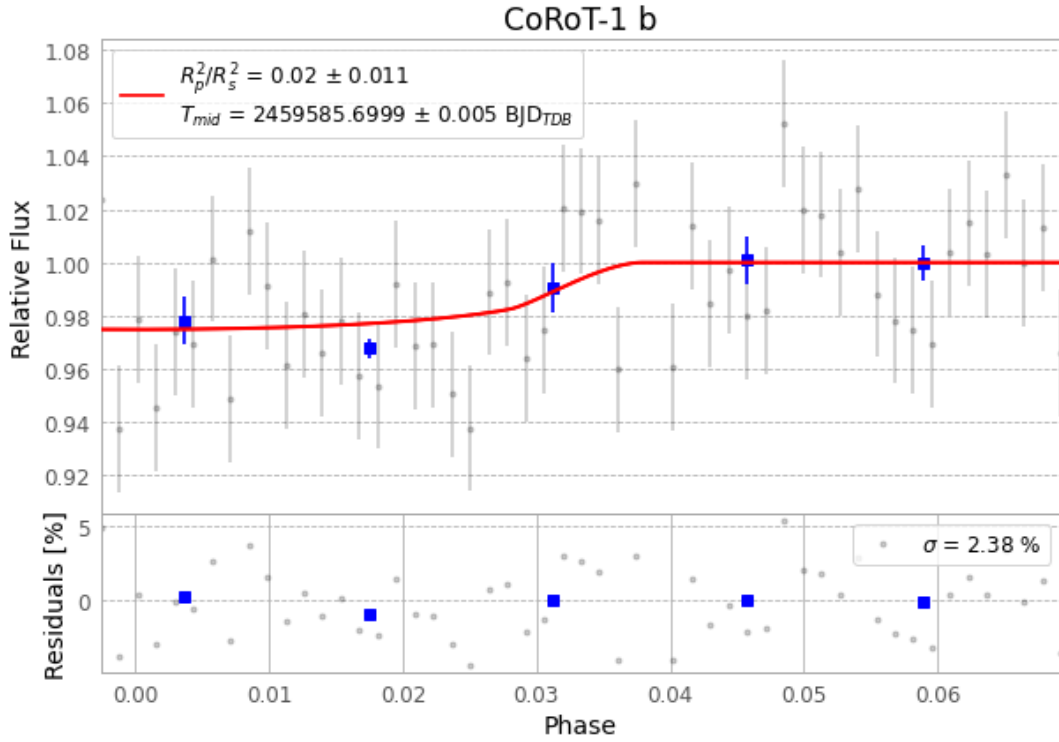


Figure 1 — FinalLightCurve_CoRoT-1 b_2022-01-05.png (SHA-256 21ea857bf336f332...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b423271e5c842189...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), CoRoT-1220106043312.FITS → CoRoT-1220106070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-220106.log (2ac10188a552...), FinalLightCurve_CoRoT-1 b_2022-01-05.png (21ea857bf336...), FinalLightCurve_CoRoT-1 b_2022-01-05.csv (ee34ce7bb9cd...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:20Z.